Inorganic Cumulative Exam
September 26, 2003

A fundamental knowledge of the synthesis of a variety of small molecules is invaluable to a chemists. For example, they may be used to prepare isotopically labeled materials from commercial sources or they may be used to generate a compound in a vacuum chamber or under other restricted circumstances. Provide reactions for the practical laboratory synthesis of the following species. Note: Simple combination of the constituent elements is rarely a practical method of synthesis. While many gases are available from cylinders, these would not be appropriate sources if a procedure was modified for isotopic labeling and therefore are not appropriate answers.

1. HCl e.g. $NaCl + H_2SO_4$ --------$HCl + Na_2SO_4$ reaction proceeds upon heating, dry HCl is produced with no clean up necessary because H_2SO_4 is not volatile under moderate heating.

2. N_2

3. O_2

4. CO_2

5. D_2

6. NH_3

7. CO

8. NO

9. N_2O

10. NO_2

11. O_3

12. Cl_2

13. SO_3

14. H_2O

15. B_2H_6